

# AI & WORKFORCE ANALYTICS

## Excel Dashboard Project Report

### 1. Introduction

AI and workforce analytics have become important tools for understanding how artificial intelligence is influencing employees, job roles, salaries, work patterns, and skill development. Organizations can use workforce data to identify salary trends, analyze AI adoption, understand remote-working patterns, and determine the skills employees may need to develop in the future.

This project, titled “**AI & Workforce Analytics,**” presents an interactive Excel dashboard designed to analyze workforce-related data using different parameters such as experience level, salary currency, AI tool usage, job roles, team size, remote work ratio, and interviews-to-offer information.

The dashboard converts raw workforce data into meaningful visualizations and key performance indicators (KPIs), making it easier to understand important trends and support data-driven decision-making.

### 2. Objectives of the Project

The main objectives of this project are:

1. To analyze average and highest salaries across the workforce.
2. To examine the relationship between job roles and AI tool usage.
3. To analyze employee experience levels and workforce characteristics.
4. To understand remote-working patterns across different industries.
5. To analyze team sizes associated with different job roles.
6. To examine interviews-to-offer trends across different salary currencies.
7. To understand the importance of upskilling in an AI-driven workplace.
8. To create an interactive and visually appealing Excel dashboard for workforce analysis.
9. To present complex workforce data in a simple and understandable format.

### 3. Tools and Technologies Used

The project was developed using **Microsoft Excel** and its data analysis and visualization features.

The major Excel features used include:

- Data cleaning and organization
- Pivot Tables
- Pivot Charts
- Dashboard design
- KPI cards
- Filters/Slicers
- Bar charts
- Pie charts
- Data aggregation
- Interactive data analysis

The dashboard provides filters for parameters such as **experience level, salary currency, AI tool usage, and switched jobs recently**, allowing users to analyze different segments of the workforce.

## 4. Dashboard Overview

The dashboard is titled “**AI & WORKFORCE ANALYTICS.**” It contains several KPI cards and charts that summarize important workforce information.

The main KPI indicators shown on the dashboard are:

| KPI                          | Value     |
|------------------------------|-----------|
| Average Salary               | 89,362.06 |
| Average Weekly Working Hours | 45.23     |
| Average AI Usage             | 6.91      |
| Highest Salary               | 276,910   |
| Average Upskill Score        | 9.51      |

These indicators provide a quick overview of salary levels, working hours, AI adoption, maximum salary, and the workforce's upskilling level.

## 5. Key Dashboard Components

### 5.1 Average Salary

The dashboard shows an **average salary of 89,362.06**. This provides an overall indication of the earning level represented in the dataset.

Salary can vary significantly depending on factors such as experience, job position, industry, location, and technical skills. Therefore, the average salary serves as a useful starting point for comparing workforce groups.

## 5.2 Average Weekly Working Hours

The average weekly working time shown in the dashboard is **45.23 hours**.

This metric helps in understanding the general workload represented in the dataset. It can also be compared with different job roles and industries to identify areas where employees may be working longer hours.

## 5.3 Average AI Usage

The dashboard reports an **average AI usage score of 6.91**.

This indicates that AI tools have a meaningful presence in the workforce represented by the dataset. The AI usage chart also demonstrates that adoption differs considerably between job roles.

## 5.4 Highest Salary

The highest salary displayed is **276,910**.

The difference between the average salary and highest salary demonstrates that compensation can vary substantially across job positions and levels of expertise.

## 5.5 Average Upskill Score

The dashboard shows an **average upskill score of 9.51**.

Upskilling is particularly important in an environment where AI and automation are changing the nature of jobs. A high focus on skill development can help employees adapt to changing workplace requirements.

# 6. Interviews-to-Offer Analysis

The **Interviews to Offer** chart compares the number of interviews and offers across different salary currencies.

The chart includes currencies such as:

- AUD
- BRL

- CAD
- DE
- FR
- GBP
- INR
- NL
- PLN
- SGD
- USD

The visualization shows considerable differences between the categories. The **USD category has the highest value**, while several other currencies have comparatively lower values.

This analysis can help organizations understand differences in recruitment activity across different markets and salary currencies.

## 7. Remote Work Analysis

The **Remote Ratio** pie chart presents remote-working patterns across different industries.

The industries represented include:

- Consulting
- Education
- Energy
- Finance
- Government
- Healthcare
- Manufacturing
- Media
- Retail

The chart indicates that remote work is not distributed equally among all industries. Some sectors have considerably higher remote-work representation, while others have lower levels.

This difference may be influenced by the nature of work. For example, jobs that depend heavily on physical locations or equipment may have fewer remote opportunities, whereas knowledge-based jobs may be more suitable for remote working.

## 8. Team Size Analysis

The **Team Size** chart compares team sizes across various job positions.

The roles represented include:

- Research Scientist
- MLOps Engineer
- Machine Learning Engineer
- LLM Engineer
- Data Scientist
- Data Science Manager
- Data Engineer
- Data Analyst
- Computer Vision Engineer
- Business Intelligence Analyst
- Analytics Engineer
- AI Engineer

The chart shows that team sizes vary considerably between roles. **Machine Learning Engineer, Data Engineer, and AI Engineer-related positions show relatively larger team sizes** compared with several other roles.

This information can help organizations understand workforce structures and determine how different AI and data functions are organized.

## 9. AI Usage by Job Role

The **AI Usage** chart is one of the important components of the dashboard.

It compares AI usage across different job positions, including:

- AI Engineer
- Analytics Engineer
- Business Intelligence Analyst
- Computer Vision Engineer
- Data Analyst
- Data Engineer
- Data Science Manager
- Data Scientist
- Machine Learning Engineer
- MLOps Engineer
- Research Scientist

The chart demonstrates that AI usage differs significantly across job roles. Some AI and machine-learning-related positions show considerably higher usage levels than traditional analytical positions.

This suggests that the extent of AI adoption is strongly influenced by the nature of the job and the technology requirements associated with the role.

## 10. Importance of Upskilling

The dashboard highlights the growing importance of workforce upskilling.

As AI becomes increasingly integrated into organizations, employees may need to develop skills in areas such as:

- Artificial intelligence
- Machine learning
- Data analytics
- Python
- SQL
- Data visualization
- Cloud technologies
- Generative AI
- Automation
- Business intelligence

The **average upskill score of 9.51** indicates that skill development is an important aspect of the workforce data represented in the project.

Organizations can use such analysis to identify skill gaps and design appropriate employee training programs.

## 11. Major Findings

The major findings from the dashboard are:

1. The overall average salary is **89,362.06**, while the highest recorded salary is **276,910**.
2. Employees work an average of approximately **45.23 hours per week**.
3. AI usage varies significantly between different job roles.
4. AI-related positions generally show stronger AI usage compared with some traditional analytical roles.
5. Remote-working patterns differ across industries.
6. Team sizes vary considerably depending on job position.
7. Recruitment activity differs across salary currencies, with the **USD category showing the highest value in the Interviews-to-Offer chart**.
8. Upskilling is an important workforce requirement in an AI-driven environment.

9. Workforce analytics can help organizations understand salary, recruitment, work arrangements, AI adoption, and skill development together rather than analyzing them separately.